

TORSIONVERSE: ACOUSTICS AND GEOMETRIC MEDIUM DEFORMATION

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Theory paper: retrodictions of Chladni, levitation, ringing rocks, stone circles; predictions of ϕ^2 coupling and N-ring dependence. No personal measurements; all claims are mechanistic derivations or predictions from the I_h character table and torsionverse framework.

Companion script: analysis/demos/acoustics_doc.py (27/27 PASS, AC1-AC15, SP1-SP7c)

Continuation: docs/doc_series2_piezo_escalator.txt (piezo + electron escalator companion paper)

Branches from: doc_orbit_pressure (pressure-following mechanism),

doc_electron (E+ vertex mode, pressure-following),
doc_jobson_cell (I_h geometry, Maxwell criticality)

ABSTRACT

Matter follows pressure gradients in the torsion medium. This single mechanism -- the same one that produces gravity and electromagnetism -- also explains acoustic phenomena at human scales. Chladni patterns, acoustic levitation, and natural geometric resonance in rocks and structures are all manifestations of the same T_{1g} pressure-following law. These are not analogies: they ARE the torsion medium mechanism operating at different scales.

Certain geometric configurations in nature -- crystal structures, enclosed cavities, ring arrangements, and apex shapes -- focus T_{1g} pressure and create low-pressure zones. Objects in those zones experience reduced effective gravity. This may explain anomalous lightness in ringing rocks, acoustic phenomena at megalithic stone circles, and levitation effects in natural vortices.

1. CHLADNI PATTERNS: DIRECT EVIDENCE OF PRESSURE-FOLLOWING

When a plate is driven at resonance, it forms standing waves. Sand or rice grains sprinkled on the plate migrate to the nodal lines (pressure nodes of the standing wave) and form the famous Chladni figures.

TORSIONVERSE MECHANISM: Sand grains drift to pressure nodes by dwell-time asymmetry -- the same mechanism that produces gravity. Near a pressure node, the wave spends more time (longer dwell time) per unit displacement; grains accumulate where dwell time is maximum. This is not radiation pressure (which pushes grains to pressure antinodes); it is the torsionverse pressure-following mechanism, which moves matter to pressure MINIMA.

Note: Standard acoustic radiation pressure theory predicts grains move to

pressure antinodes for objects larger than the wavelength/2, and to nodes for smaller objects. The torsionverse provides the underlying mechanism for the "smaller object" case (dwell-time asymmetry) and connects it to gravity and EM through the same pressure-following law.

A_g ORTHOGONAL ENHANCEMENT:

Two acoustic sources crossed at 72 degrees (C5 icosahedral angle) produce more A_g mode excitation at the crossing by factor $\phi^2 = 2.618$ vs 90 degrees.

Prediction: Chladni patterns driven by two sources at 72 degrees should show more defined and symmetric patterns than at 90 degrees, with ϕ^2 more visible grain concentration at the crossing point.

Experiment: two small speakers, adjustable angle bracket, sand on a flat plate, vary crossing angle from 0 to 180 degrees in 12-degree steps.

Cost: < \$20. [See experiments/bench_test_orthogonal_waves.txt for hardware]

2. ACOUSTIC LEVITATION

Acoustic levitation uses standing waves to hold small objects in mid-air at pressure nodes. Standard explanation: radiation pressure balances gravity.

TORSIONVERSE MECHANISM: Objects drift to pressure minima by dwell-time asymmetry. In a vertical standing wave, the pressure nodes are fixed points where the wave is stationary. Objects accumulate there because their local dwell-time gradient is zero (stable equilibrium) -- the same reason a planetary orbit is stable.

TORSIONVERSE PREDICTION:

(a) The levitation force scales with T_1g coupling efficiency of the medium

at the crossing angle. A 72-deg crossing geometry should levitate objects with $\phi^2 = 2.618$ x less required acoustic power than a 90-deg geometry.

(b) Objects aligned with icosahedral cell axes (as set by local proton

chirality) will levitate with lower required field than randomly oriented objects.

OPEN: Compute the Bjoerknes radiation force from torsionverse parameters

(G_shear, rho, Rs) and compare to measured values.

3. NATURAL GEOMETRIES THAT CREATE LOW-PRESSURE ZONES

The coupling strength at the focus of any geometry is given by $\chi(T_{1g})$

evaluated at the dominant rotation angle (I_h character table, Cotton 1990):

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chi(T_1g, 72 deg, C5) = phi = 1.618 [maximum in I_h]
chi(T_1g, 120 deg, C3) = 0 [exact null]
chi(T_1g, 180 deg, C2) = -1 [destructive]
N-ring (adjacent spacing): chi = 1 + 2*cos(360/N)
N=3 -> 0 | N=5 -> phi | N=10 -> phi^2 | N->inf -> 3 (full circle)
N=4 RING (adjacent 90 deg): chi = +1 (positive).
4-sided PYRAMID (opposite faces 180 deg): chi = -1.
Same count, different dominant angle, different sign.
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Any geometric arrangement that focuses T_{1g} (pressure) mode waves to a region creates a pressure minimum there. Coupling efficiency at the focus depends on the symmetry of the arrangement via $\chi(T_{1g})$ at the relevant rotation angle.

Objects in the low-pressure zone experience reduced effective gravity. The engineered pyramid is one instance; natural equivalents exist across many scales.

3.1 CRYSTAL RESONANCE IN ROCKS -- THE RINGING STONES MECHANISM

Some rocks produce a clear musical tone when struck ("ringing rocks," "singing stones"): Ringing Rocks Park (Pennsylvania), bluestones at Stonehenge, various megalithic sites globally. Standard geology notes these tend to be high-silica rocks with olivine or pyroxene inclusions.

TORSIONVERSE MECHANISM:

- The rock's size and crystal domain structure set its resonance frequencies
- Olivine/pyroxene inclusions have approximate icosahedral packing in their atomic coordination shells $\rightarrow T_{1g}$ mode couples at those frequencies
- Seismic background noise (Schumann resonances 7-45 Hz, microseisms 0.1-0.3 Hz) continuously drives the rock at its natural resonance
- At resonance: internal standing wave creates a T_{1g} pressure pattern
- Pressure minimum inside $\rightarrow$ the rock is lighter than a non-resonant rock of the same material and size

PREDICTIONS:

- (a) Ringing rocks are measurably lighter per unit volume than non-ringing rocks of the same parent material and size. Weight deficit $\sim Q$ factor of the resonance.
- (b) A ringing rock driven at its resonance by an external acoustic source shows measurable apparent weight reduction on a precision scale. Effect is fully reversible when driving stops (dynamic eddy, not a T_{2g} vortex; relaxes on acoustic timescale, not $\tau_{\text{relax}} = 10^8$ s).
- (c) Megalithic builders may have selected ringing stones partly because they were lighter to transport (Stonehenge bluestones chosen from Preseli Hills, Wales -- these ring when struck; the 250 km transport then makes more sense).

3.2 APEX GEOMETRIES IN NATURE

Any tapered shape focuses arriving T_{1g} waves to its tip, creating a pressure minimum at the apex. Crystal symmetry of the material determines coupling efficiency:

- Quartz crystal terminations: hexagonal prism + rhombohedral cap (C_3 symmetry). C_3 not in $I_h \rightarrow \chi(T_{1g}, C_3) = 0$. Zero crystal coupling; geometric focus only.
- Calcite stalagmites / stalactites: trigonal (C_3) symmetry. Same: zero crystal coupling, geometric apex effect only.
- Olivine (orthorhombic, C_{2v}): some C_2 coupling: $\chi(T_{1g}, C_2) = -1$. Slight destructive coupling. Olivine-bearing tapers are marginally worse than amorphous.
- Pyrite (FeS_2): despite pseudo-icosahedral reputation, crystal symmetry is O_h (cubic), not I_h . C_4/C_2 crossing $\rightarrow \chi(T_{1g}, C_2) = -1$. Suboptimal.

NOTE: No common natural crystal achieves C_5 coupling in a single-crystal pyramid. C_5 coupling requires I_h geometry -- the Jobson cell itself. Natural materials approximate C_5 only in disordered or quasicrystalline forms (Icosahedrite, Al-Mn quasicrystal).

3.3 ENCLOSED CAVITIES: GEODES AND STONE CHAMBERS

A closed or near-closed cavity creates internal standing waves at its resonant frequencies. At those frequencies, the center of a spherical or dome-shaped cavity becomes a T_{1g} pressure minimum -- the acoustic analogue of a Fabry-Perot resonator.

- Geodes (hollow mineral nodules): roughly spherical quartz or calcite shell. At resonance, the interior center is a pressure minimum. A geode at acoustic resonance is lighter than the same geode off-resonance.
- Corbelled stone chambers (Newgrange, Irish passage tombs, Mycenaean tholos tombs): domed interior has a well-defined acoustic resonance. Seismic driving creates a chronic low-pressure zone at the focal center. This may be the functional basis for the dome geometry. Computed for Newgrange (R~3 m chamber): $f = v/2R = 57$ Hz (spherical approx); actual corbelled geometry has $R_{\text{eff}} \sim 1.56$ m giving $f \sim 110$ Hz (consistent with published archaeoacoustics; Watson & Keating 1999). Helmholtz mode of the 19 m passage + chamber: $f = 1.18$ Hz (deep infrasound).
- Helmholtz resonator caves: large cave interior with narrow entrance neck. At $f = (v/2\pi) \sqrt{A/(L \cdot V)}$, interior pressure is a minimum relative to outside. Objects inside experience slightly reduced effective gravity at that frequency.

3.4 RING GEOMETRIES: STONE CIRCLES AS STATIC CAROUSELS

A ring of N upright stones with vertical long axes acts as the static version of the spinning N-pyramid carousel. Driving is seismic and acoustic background rather than centrifugal. N-dependence is identical:

N=3 (triangle of 3 stones): $\chi(T_{1g}, C3) = 0$ -- null at center
 N=4 (square of 4 stones): $\chi(T_{1g}, C2) = -1$ -- destructive at center
 N=5 (pentagon of 5 stones): $\chi(T_{1g}, C5) = \phi$ -- maximum at center

STONEHENGE (horseshoe of 5 trilithons): approximate C5 symmetry, opening toward midsummer sunrise. The altar stone sits at the focal point -- where the T_{1g} pressure minimum would be for a C5 ring. If intentional, this is the optimal ring configuration from I_h character theory.

PREDICTION: Stone circles with N=5 (or N=10, 15) show stronger documented acoustic and anomalous phenomena at their centers than N=3 or N=4 circles.

3.5 VORTEX GEOMETRIES

A rotational fluid vortex has a low-pressure axis. In the torsionverse, a sustained vortex is a T_{2g} (shear/magnetic) mode excitation -- distinct from the T_{1g} geometric focus of the geometries above.

- Tornado eye: very low atmospheric pressure, low wind at center. Reports of objects rising inside tornado eyes may include a T_{2g} component. T_{2g} vortex $\tau_{\text{relax}} \geq 10^8$ s in matter-free voids -- the eye can sustain reduced effective gravity along its axis beyond what aerodynamics explains.
- Whirlpool center: same mechanism at smaller scale.

NOTE: T_{2g} (vortex) and T_{1g} (geometric focus) are complementary but distinct.

Pyramids, rings, and cavities operate by T_{1g} focusing. Vortices operate by T_{2g} shear depletion. Both create low-pressure on axis; different geometries produce them.

4. OPEN ITEMS

- Ringing rock density anomaly: weigh ringing vs non-ringing rocks of same type and size; test whether lighter correlates with higher Q factor
- Ringing rock driving test: drive a ringing rock at its resonance with a speaker; measure apparent weight reduction on a precision scale
- Stone circle survey: tabulate N values globally; test whether documented center phenomena correlate with $N > 5$ (more stones = more coupling)
- Cave Helmholtz: identify natural caves with well-defined neck-cavity geometry; measure interior pressure at Helmholtz frequency
- Geode driven resonance: seal small geode, drive at resonance, measure apparent weight change on precision scale
- Quasicrystal anomalous coupling: check published measurements on Al-Mn and other icosahedral quasicrystals for anomalous EM or acoustic resonances
- Kozyrev mirror connection: if the focal geometry of a curved mirror creates a T_{1g} pressure minimum equivalent to a ring focal point, the mechanisms unify
- Bjoerknes radiation force: compute from torsionverse parameters (G_{shear} , ρ , R_s) and compare to published levitation force measurements